

Dalena Nguyen

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EDUCATION

University of Florida (GPA 3.86)

- Major: Computer Science | Minor: Digital Arts and Sciences

August 2022 - May 2025

Gainesville, FL

St. Petersburg College (GPA 4.0)

- Associates of Arts Degree

August 2021 - May 2022

Tarpon Springs, FL

EXPERIENCE

United States Air Force

May 2024 – August 2024

Full Stack Developer Intern (T.S Clearance)

Eglin Air Force Base, FL

- Developed microservice functions for serial RS-232/422/485 protocols, containerized the application using Docker, and managed deployment with Kubernetes.
- Enabled automated, repeatable, and distributable tests to assess firmware/software resiliency and detect vulnerabilities or potential exploits of embedded weapon systems.

PROJECTS

Filesystem in User Space (C++)

- Developed a FUSE-based filesystem daemon to enable read and write access to WAD data files.
- Allows users secure, convenient, non-privileged mounts, reducing risks of kernel crashes.

Memory Manager (C++)

- Developed an efficient memory manager to handle allocation and deallocation, minimizing memory fragmentation and optimizing memory usage by 20%.
- Integrated custom memory tracking algorithms to prevent leaks and maintain data integrity.

Playlist Generator (C++)

- Designed a playlist generator using Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms, improving user control over playlists compared to traditional platforms like Spotify.
- Integrated a dynamic SMFL frontend allowing customization based on genre, energy level.

REACT Website (JavaScript, HTML)

- Developed a dynamic and responsive personal website using React to showcase portfolio projects, technical skills, and professional experience.
- Integrated a clean and user-friendly UI with React components, enabling easy navigation and browsing.

LEADERSHIP/INVOLVEMENT

Women in Computer Science

August 2022 - Current

- Mentored new members, helping them develop technical and networking skills.

Society of Women Engineers (STEAM Team)

August 2023 – May 2024

- Led a team in designing and implementing an interactive exhibit using Arduino Nano, light sensors, and motors, attracting over 200 visitors at the CADE Museum for Creativity and Innovation.
- Organized cross-functional team meetings to troubleshoot technical challenges, ensure smooth project delivery, and enhance project coordination.

Programming Languages/Tools: C++, Python, Git, JavaScript, Java, Docker, Linux, Unity (C#)